

Storage referential based on iRODS, for patient data in Gustave Roussy cancer care center.

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ABSTRACT

Cancer care and clinical research produce large amounts of data when performing routine genomics profiling and digital histological slides. Clinical research projects often involve multiple data providers over time and rely on both internal and external data producers. To harmonize and centralize data management, an on-premise storage referential has been deployed at Gustave Roussy Comprehensive Cancer Center. Its primary goal is to enable data reuse for retrospective analyses and the creation of custom patient cohorts while fostering scientific collaborations. The storage referential consists of two main components: (1) a dedicated data catalog that manages storage units and their standardized metadata descriptions, and (2) an iRODS-based file system.

The data catalog implements data lifecycle management, ensuring appropriate access rights and full traceability through standardized metadata descriptions based on controlled vocabularies.

Metadata and user rights in the iRODS file system are not directly exposed to end users; they are automatically inherited from the data catalog.

Dedicated data management routines automate recurring tasks and enforce business rules specific to each context, including decisions on which data formats should be retained over time. Patient and sample identifiers, captured as metadata, are central to reflecting data availability and increasing the overall value of the storage referential.

Since its deployment, the storage referential has been in production for six years. It specializes in raw data storage, guarantees the absence of file duplicates across the infrastructure, and is used daily by computational specialists for both routine diagnostics and retrospective research.

Keywords

Data Management, Cancer Care, Clinical Research, Interoperability, FAIR Data.

INTRODUCTION

As France's and Europe's leading comprehensive cancer center, and the world's sixth best oncology hospital according to Newsweek, Gustave Roussy is a world-class institute dedicated to cancer care, research, and education. It treats patients with all types of cancer, at every age and every stage of the disease. Its expertise in rare cancers and complex tumours is internationally recognized. Gustave Roussy brings together medical, scientific, and technological excellence.

Advances in next-generation sequencing technologies and digital pathology have profoundly transformed cancer research and clinical practice. Molecular profiling and high-resolution imaging now generate terabytes of data across diagnostic and research activities. These data are subsequently used for retrospective analyses such as digital cohorts (for example [1]), reproducibility studies, and the development of future scientific projects.

Managing such volumes of data presents significant challenges. Data files are often produced by multiple internal and external providers, generated by different technologies at various time points, and associated with complex clinical contexts. To address these challenges, Gustave Roussy has implemented an institutional storage referential designed to centralize all data while preserving the flexibility required for project-specific analytical needs.

This storage solution combines a metadata catalogue with an iRODS-based storage infrastructure. Its objective is to ensure long-term preservation, facilitate data traceability, promote reuse and guarantee secure accessibility.

In the context of clinical research, genomics profiling and digital histological slides are routinely produced. These represent large volumes of data generated by various teams and external partners, which must remain available for complex analyses on high-performance computing (HPC) infrastructures.

STORAGE REFERENTIAL COMPONENTS

The data catalog is a platform originally developed by Institut Curie [2]. It serves as the reference source for controlled metadata vocabularies, data lifecycle management, and access rights.

The fundamental entity in the system is the storage unit, which relies on a barcoding system. Storage units are data-agnostic and identified by a unique Uniform Resource Identifier (URI) called a dataset, implemented as an iRODS collection. These datasets can be referenced unambiguously across projects and support reproducible scientific workflows.

A storage unit is defined by the following attributes: a biological sample, a production time point, a modality of data acquisition, Project affiliation.

For example, a patient undergoing normal tissue whole-exome sequencing (WES), tumor WES, and tumor RNA sequencing (RNA-seq) generates three independent storage units. If the same acquisition (e.g., RNA-seq) is repeated at a later time point, a new storage unit is added.

Storage units can also be linked together to represent relationships between datasets. Typical examples include:

- Linking sequencing capture-kit reference files (e.g., WES kit genomic coordinate files or BED files) to the corresponding generated sequencing datasets.
- Linking one or multiple raw data source files to computed data storage units.

Computed data storage units differ from raw data storage units in that they can correspond to multiple patients, samples, or time points. Users can register a Git repository and commit hash as an analysis-type attribute of a computed data storage unit to ensure full traceability of the code version that produced the results. When coupled with container technologies such as Docker or Singularity on the HPC infrastructure, this approach strongly supports scientific reproducibility.

Primary dataset INPUT:

```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [],
  "protocolName": "Exome-seq",
  "tpName": "NovaSeq X Plus",
  "datasetName": "S17765_1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_PROJECT"
}
```

Analysis dataset INPUT:

```
{
  "organisationName": "Gustave Roussy",
  "datasetIds": [<primary datasetId>],
  "analysisTypeName": "https://gitlab.com##<commit-hash>",
  "datasetName": "analysis1",
  "biologicalCategoryName": "genomics",
  "projectName": "GR_RETROSPECITE_PROJECT"
}
```

OUTPUT is a storage unit identifier called `datasetId`, that can be used to build URI:

```
/odin/kdi/dataset/<datasetId>
```

The catalog stores standardized metadata descriptions in JSON format, using controlled vocabularies. These metadata are interoperable with institutional patient and sample reference systems through key/value mechanisms called aliases. Key metadata elements include patient and sample identifiers, which are linked to the corresponding storage unit. This already provides rich context regarding data production.

Although stored datasets may be heterogeneous, metadata standardization offers a consistent framework for data management across the entire infrastructure. Computational pipelines remain responsible for handling project-specific format diversity.

Sample declaration template:

```
{
  "sampleType": "tumor",
  "speciesName": "Homo sapiens",
  "sampleNameType": "Sample number",
  "sampleName": "S12345",
  "sampleAliases": [],
  "comment": "",
  "patientGender": "female",
  "patientNameType": "NOIGR",
  "patientName": "PatientName",
  "patientAliases": [
    {
      "typeName": "Patient inclusion number",
      "value": "P12345"
    }
  ],
}
```

Access control is managed exclusively through the data catalog. Two principal user group categories are defined:

- Read-only groups, corresponding to scientific projects, clinical trials, or research programs (primarily used by bioinformaticians).
- Write-access groups, dedicated to data managers.

Only datasets with read-only status are visible to end users. This lifecycle-based approach prevents access to incomplete or provisional data while ensuring that released datasets remain immutable and fully traceable. Metadata and access policies are automatically propagated to the iRODS infrastructure at the storage unit level.

As an example,

Lifecycle step1: 2000016 is a storage unit corresponding to an iRODS collection owned by admin account. It is also owned by write access iRODS user group. Reader group configured in metadata catalog are not yet propagated to iRODS:

```
>ils -A /odin/kdi/dataset/2000016/archive
/odin/kdi/dataset/2000016/archive:
    ACL - g:systemGroup1#odin:own
          SVC_ODIN@INTRA.IGR.FR#odin:own
```

Lifecycle step2: 2000016 is now read-only storage unit corresponding to an iRODS collection owned by admin account and the read access is configured with iRODS user group:

```
>ils -A /odin/kdi/dataset/2000016/archive
/odin/kdi/dataset/2000016/archive:
    ACL - g:MATCH-R-ap#odin:read object
          g:MR_immuno_lung-ap#odin:read object
          g:MetaPRISM-ap#odin:read object
          g:systemGroup1#odin:read object
          SVC_ODIN@INTRA.IGR.FR#odin:own
```

The data catalog is implemented as a MySQL relational database and is managed by a Java Spring Boot backend application exposed as a REST API. The Java iRODS library irods4j [3] is used with an iRODS administrator account to automatically propagate both access rights and metadata from the data catalog.

iRODS instance on premise

The iRODS platform is deployed on-premise using a containerized Docker-based architecture. The deployment includes an iRODS server connected to a PostgreSQL database and integrated with the institutional LDAP directory service for centralized user authentication. Data is stored on-premise using Dell/EMC Isilon storage infrastructure.

To facilitate metadata management, the data catalog is replicated within iRODS using the imeta framework. This approach enables metadata-based searches directly in iRODS while maintaining consistency with the primary catalog. iRODS metadata are read-only for end users.

Additionally, iRODS collections are organized into barcoded storage units. This results in a flattened collection hierarchy that reflects the underlying physical storage organization and simplifies data management operations.

For example, users can query the controlled metadata vocabulary directly in iRODS. Note that extended metadata, such as patient and sample identifiers, are not displayed here.

```
iquery "select META_COLL_ATTR_NAME,META_COLL_ATTR_VALUE,META_COLL_ATTR_UNITS WHERE
COLL_NAME = '/odin/kdi-dev/dataset/2123410'"
```

iRODS attribue	iRODS value	iRODS unit	comment
projectName	Routine_Hematology		Primary project
datasetArchived	true		Storage unit is read-only if true
datasetCreationDate	2024-04-27T14:30:36Z		Storage unit creation date
datasetName	240426_M02317_0900_000000000-L7VL3		For data produced by Gustave Roussy core facilities, the barcode of the sequencing run is reported.
projectName	mooseQC_BIOPATH		Other project
organisation	Department of Biology and Medical Pathology		Team that produces the data
dependsOn	2081284	datasetId	Logical link between storage unit
technology	sequencing		Describe the files stored in the current storage unit.
protocol	Target-seq		
technicalPlatform	MiSeq		
biologicalCategory	genomics		

Table 1. minimal metadata controlled vocabulary in iRODS from data catalog.

STORAGE REFENTIAL ALIMENTATION

As previously mentioned, iRODS primarily serves as a secure storage backend. Metadata governance and user permissions are fully delegated to the data catalog, ensuring clear separation between storage and governance responsibilities.

Populating the storage referential requires two main steps: 1/Data declaration 2/Data import.

Data ingestion begins with the declaration of a storage unit through the data catalog REST API. This creates the corresponding iRODS collection and automatically assigns the associated metadata and access policies.

A key design principle of the storage referential is its ability to accept data with minimal metadata descriptions. This approach encourages systematic use of the platform, even when complete biological or clinical annotations are not yet available.

Data managers subsequently enrich the metadata as additional information becomes available, particularly patient and sample identifiers. This progressive enrichment balances operational efficiency with long-term metadata quality.

Data files are then imported into the designated storage unit using iRODS transfer commands with checksum verification enabled (`iput -K`). iRODS natively maintains checksum information for stored files. All checksums across the infrastructure (managed via the iRODS admin account) are accessible through a REST API. Additionally, a dedicated `checkAndDestroy` routine allows users to verify that files have been safely stored as read-only within iRODS before deleting local working copies. It reduces storage pressure on computational infrastructure while maintaining full data availability.

Example of remote check from a locally computed checksum:

```
>curl --request GET '<endpoint>/datafile/existsChecksum' --data
"96fa55c907283a75ec3f6ca6384cced8"
[{"dataPath":"/odin/kdi/dataset/2000016/archive/file_R1.fastq.gz","dataSize":"21698235
94","checksum":"96fa55c907283a75ec3f6ca6384cced8"}]
```

For recurrent workflows, automated agents have been developed as a Java Spring Boot executable JAR file. These agents provide the following functionalities:

- Automated storage unit creation
- Metadata templating
- File import orchestration
- Error recovery mechanisms

The agents are designed to be resilient. Failed imports automatically trigger retry mechanisms, while checksum verification prevents unnecessary duplication of files already present in the infrastructure.

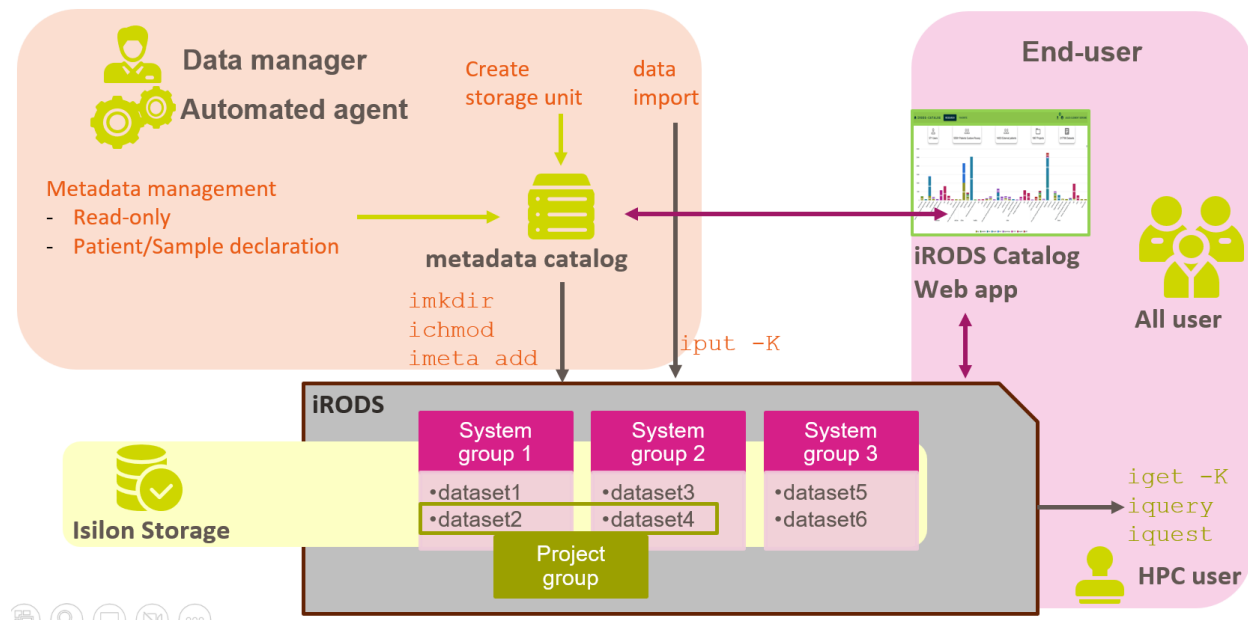


Figure 1. Storage Referential: iRODS-based system

DATA MANAGER ADMIN TASK

As part of Data Manager administration tasks, a verification process was implemented to ensure that no duplicate files, identified by identical checksums, exist within the iRODS environment. To support this validation, a script was developed to consolidate the output of iRODS queries (`iquest`), generating a comprehensive inventory of all data objects stored in the system along with their associated checksums. The resulting dataset is analyzed to detect potential checksum duplicates, enabling the identification of redundant files and contributing to the overall integrity and consistency of the data repository.

Interrupted imports or workflow failures may leave empty storage units in the catalog. Dedicated administrative procedures identify and remove these orphan entities, thereby preserving catalog consistency and reducing metadata clutter.

As research projects evolve, the metadata structure occasionally requires corrections or contextual adaptations. Data managers perform targeted updates based on user feedback and project requirements. Regular reporting activities provide project-specific summaries that support governance and quality monitoring.

STORAGE REFERENTIAL END USER ACCESS

Data access

Most specialist users access the storage referential directly from the institutional high-performance computing (HPC) environments.

Common operations include:

- Metadata search using `iquery`
- Advanced queries using `iquest`
- Data retrieval using `iget -K`

Following analysis completion, users are encouraged to remove local copies through the `checkAndDestroy` workflow to minimize HPC storage consumption.

An internal web portal, called "irods-catalog", was developed to provide a graphical user interface (GUI) for visualizing the content of the storage referential.

Through this interface, users can:

- View a summary of the storage referential content
- Browse storage units
- Explore metadata
- Search available datasets using metadata filters
- Download authorized files

This interface broadens accessibility beyond users familiar with command-line iRODS tools and enables interoperability with other applications that reference storage units.

Data managers retain direct access to advanced iRODS querying capabilities and can perform on-demand investigations using iCommands from the Unix shell when required.

The screenshot shows the iRODS-CATALOG internal web portal. At the top, there is a green header bar with the text "IRODS - CATALOG" and buttons for "RESEARCH" and "FAVORITE". On the right, there is a user profile for "JULES-CLEMENT GEROME". Below the header, there is a navigation bar with "5 DATASET" and "4 PATIENT" tabs. To the right of the tabs are buttons for "COLUMNS" and "EXPORT AS CSV", and a "Rows per page: 10" dropdown menu. The main content is a table with the following columns: DatasetId, Import date, Biological category, Organisation, Protocol, Sample type, Patient, Datafile count, Depends on id, Sample, and Project. The table contains five rows of data, each representing a dataset with its associated metadata and sample information.

DatasetId	Import date	Biological category	Organisation	Protocol	Sample type	Patient	Datafile count	Depends on id	Sample	Project
2000019	2020-04-24	genomics	IntegraGen	Exome-seq	healthy	patient_ID1 patient_aliasA	2	2000307	sample_ID10 sample_aliasZ	MATCH-R (15)
2000074	2020-04-24	genomics	IntegraGen	Exome-seq	tumor	patient_ID1 patient_aliasA	2	2000307	sample_ID90 sample_aliasG	MATCH-R (13)
2006441	2021-02-16	transcriptomics	Novogene	mRNA-seq	tumor	patient_ID2 patient_aliasB	2		sample_ID65 sample_aliasJ	MATCH-R (10)
2040410	2022-03-11	genomics	Gustave Roussy	Target-seq	tumor	patient_ID3 patient_aliasC	1	2040389	sample_ID22	SAFIR_BREAST(0)
2237545	2026-05-18	Histology_Slides	Gustave Roussy	HE	tumor	patient_ID4 patient_aliasD patient_aliasE	1		sample_ID11	MEDITWIN (0)

Figure 2. irods-catalog internal web portal.

IRODS VERSION MIGRATION STRATEGY

The platform was initially deployed in 2020 using iRODS version 4.2.7 and was upgraded to version 5.0.2 in 2026.

To minimize service disruption and avoid modifications to the existing data catalog, a parallel migration strategy was adopted.

A new iRODS 5.0.2 instance was deployed alongside the existing iRODS 4.2.7 environment, which remained in production. Both instances were configured to access the same underlying storage system, allowing the migration to proceed without data duplication.

The iRODS 4.2.7 instance remained fully operational throughout the transition period, ensuring uninterrupted access to datasets. This also allowed progressive adaptation of data management routines for ingesting new data into the storage referential.

Existing datasets were registered in the new instance using the iRODS `ireg` mechanism, thereby avoiding physical data transfers. Metadata stored in the original environment was subsequently synchronized with the new platform to ensure consistency between the data catalog and the iRODS 5.0.2 metadata repository.

This strategy enabled a seamless migration while preserving data integrity, metadata consistency, and continuous service availability.

CONCLUSION

The increasing scale of biomedical data production requires a storage solution that goes well beyond a simple file repository.

The Gustave Roussy storage referential provides a robust and scalable solution for the long-term management of omics and digital pathology data.

After six years of production use (since 2020), the platform has proven its ability to support both routine clinical analyses and large-scale retrospective research studies. It currently includes:

- 203.272 storage units
- 1 Pb storage
- 55.000 patients followed in Gustave Roussy

Several benefits have emerged:

- Centralization of heterogeneous datasets within a unified governance framework
- Facilitation of retrospective cohort construction
- Enhanced collaboration between clinical and research teams
- Reduced operational burden through automated data management routines

Patient and sample identifiers are particularly valuable metadata elements, as they establish connections between datasets and the underlying clinical reality. These links significantly maximize the scientific value of stored data.

Most importantly, the system promotes FAIR data principles (Findable, Accessible, Interoperable, Reusable). Its architecture demonstrates how metadata standardization, automation, and strong governance can transform institutional storage systems into strategic assets for translational cancer research.

PERSPECTIVES

Future developments of the platform may include the integration of an archiving system, such as LTO9-based magnetic tapes, to support long-term data preservation and low-cost accessibility.

Additional enhancements could involve automated dataset extraction and anonymization workflows, facilitating secure data sharing with external collaborators while protecting patient confidentiality.

The introduction of automated quality control procedures, including genomic-based data pairing to ensure metadata matches the correct patient, would further improve data integrity and reliability.

Finally, hosting analysis results within the platform would provide users with streamlined access to outputs, thereby enhancing reproducibility and supporting collaborative research efforts.

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REFERENCES

- [1] Yoann Pradat, Julien Viot, Andrey A. Yurchenko, Konstantin Gunbin, Luigi Cerbone, Marc Deloger, Guillaume Grisay, Loic Verlingue, Véronique Scott, Ismael Padioleau, Leonardo Panunzi, Stefan Michiels, Antoine Hollebecque, G r me Jules-Cl ment, Laura Mezquita, Antoine Lain , Yohann Lorient, Benjamin Besse, Luc Friboulet, Fabrice Andr , Paul-Henry Courn de, Daniel Gautheret, Sergey I. Nikolaev; Integrative Pan-Cancer Genomic and Transcriptomic Analyses of Refractory Metastatic Cancer. *Cancer Discov* 1 May 2023; 13 (5): 1116–1143. <https://doi.org/10.1158/2159-8290.CD-22-0966>
- [2] Servant N, Rom jon J, Gestraud P, La Rosa P, Lucotte G, Lair S, Bernard V, Zeitouni B, Coffin F, Jules-Cl ment G, Yvon F, Lermine A, Poulet P, Liva S, Pook S, Popova T, Barette C, Prud'homme F, Dick J-G, Kamal M, Le Tourneau C, Barillot E and Hup  P (2014) Bioinformatics for precision medicine in oncology: principles and application to the SHIVA clinical trial. *Front. Genet.* 5:152. doi: <https://doi.org/10.3389/fgene.2014.00152>
- [3] <https://github.com/irods/irods4j>: A Java client library for iRODS